

Scalability & Future Plan

Date	5 July 2026
Team ID	XXXXXX
Project Name	Flood Prediction System
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application accepts manually entered weather parameters instead of real-time weather data.	Users must enter data manually, reducing automation.	High
2	The application is deployed only in a local environment.	Cannot be accessed by remote users over the internet.	High
3	The system does not include user authentication or alert notifications.	Limited functionality and user management.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local single-user access.	Deploy the application on a cloud platform (Render, AWS, or Azure) to support multiple concurrent users.
2	Data Storage	Uses a static Kaggle dataset and saved model files.	Integrate a database such as MySQL or PostgreSQL and support larger, continuously updated datasets.
3	Performance	Predictions are generated locally using the trained model.	Optimize the application using caching, model optimization, and cloud resources to improve performance under higher loads.
4	Security	No authentication or role-based access control is implemented.	Add secure user authentication, HTTPS, and role-based access to improve security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add secure user authentication, HTTPS, and role-based access to improve security.	Within 3 months	Enables live flood prediction without manual data entry.
Phase 3	Deploy the application on a cloud platform and implement user authentication.	Within 6 months	Improves accessibility, scalability, and security.

Phase 4	Develop a mobile application, add GIS-based flood mapping, and implement SMS/Email alert notifications.	Within 12 months	Expands usability, improves disaster response, and provides timely alerts to users.
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